Charpy Impact Testing and the Jominy End-Quench Test

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Group 1 (Friday Batch)

Course: UMT312L — Mechanical Testing Laboratory

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1 Objective

1. To measure the Charpy V-notch impact energy of copper, stainless steel and EN-8 plain carbon steel in three heat-treatment conditions — furnace cooled, air cooled and water quenched — at sub-zero temperature, and to express the result as a specific impact energy referred to the cross-section under the notch.
2. To measure the Vickers hardness of the three EN-8 conditions and to relate impact toughness to hardness.
3. To observe the fracture surfaces and classify them as brittle (cleavage) or ductile (fibrous), including the presence of shear lips.
4. To determine the hardenability of EN-8 by the Jominy end-quench test (ASTM A255): to measure the Rockwell hardness as a function of distance from the water-quenched end, to identify the distance at which the cooling rate becomes equivalent to air cooling, and to locate the Jominy distance for 50 % martensite.
5. To explain why the body-centred cubic steel shows a ductile-to-brittle transition while face-centred cubic copper and austenitic stainless steel do not.

2 Apparatus and Materials

- **Impact testing machine:** pendulum type, pendulum mass 21.79 kg, capacity 300 J, with both a digital (electronic) display and a mechanical energy dial. Quoted reading errors: electronic ±2.26 J, manual dial ±0.2 J.
- **Charpy specimens:** nominal $55 \times 10 \times 10$ mm with a 2 mm deep V-notch, 45° included angle, root radius 0.25 mm. Measured dimensions are given in Table 1.
- **Materials:** commercially pure copper; austenitic stainless steel; EN-8 plain carbon steel in three conditions — furnace cooled, air cooled and water quenched. The three EN-8 specimens were austenitised at 900 °C for 30 minutes before being cooled at the three rates. The set of five is the one assigned to Friday Group 1 in the handout.
- **Cooling medium:** liquid nitrogen bath. Specimen temperature was taken with an infrared thermometer immediately before and after the test.

- **Hardness testers:** Vickers machine, 136° diamond pyramid, 100 kgf, 10 s dwell, with an eyepiece micrometer for the two diagonals; Rockwell machine for the Jominy bar, C scale (diamond brale, 150 kgf).
- **Jominy equipment:** muffle furnace, end-quench fixture with a controlled water jet, and a surface grinder for the flat.
- **Jominy specimen:** EN-8 cylinder, nominally 102 mm long and 25.4 mm diameter, austenitised at 900 °C for 30 minutes — the schedule the handout assigns to Friday Group 1.
- Vernier calliper for specimen dimensions.

3 Theory

3.1 What the Charpy test measures

A notched bar is broken by a swinging pendulum of mass m released from a height h_1. The pendulum rises to a height h_2 on the far side, and the energy absorbed in breaking the specimen is

$$KV = mg\,(h_1 - h_2). \tag{1}$$

This energy is not a material constant. It contains the elastic and plastic work done in the specimen, the work of creating the two fracture surfaces, and the kinetic energy carried away by the two broken halves. The test is therefore a *comparative* one: it ranks materials and conditions tested on the same machine with the same specimen geometry, and it is fast enough to be repeated at many temperatures.

The notch is essential. It raises the local stress and, more importantly, it creates a region of triaxial tension ahead of its root. Triaxial tension raises the stress needed for plastic flow without raising the stress needed for cleavage, so a notch pushes a material towards brittle behaviour that the same material would not show when smooth.

3.2 Specific impact energy

To compare specimens whose sections differ slightly, the absorbed energy is divided by the cross-sectional area remaining under the notch:

$$a_k = \frac{KV}{A_0}, \qquad A_0 = Wb, \tag{2}$$

where b is the ligament depth left after the 2 mm notch and W is the specimen width perpendicular to it. The handout settles the geometry: the V-notch is machined *across one of the larger faces*, so it cuts into the 10 mm dimension measured perpendicular to that face, and the fracture runs over the remaining $W \times b$ rectangle. In this report $A_0 = (H - 2) \times W$ has been used, with H the height into which the notch is cut. Because both nominal dimensions are 10 mm and the measured pairs differ by at most 0.18 mm, interchanging W and H would change every a_k by under 2 % and would not alter a single conclusion.

3.3 The ductile-to-brittle transition, and why only some metals have one

In a body-centred cubic metal the stress needed to move a dislocation — the Peierls stress — rises steeply as temperature falls, because slip depends on the thermally activated formation of kinks on screw dislocations. The cleavage stress is almost independent of temperature. Below the temperature at which the yield stress crosses the cleavage stress, the material breaks before it can flow, and the absorbed energy collapses from an upper shelf to a lower shelf. EN-8 is a ferrite–pearlite steel and is body-centred cubic, so it has a transition.

In a face-centred cubic metal there are twelve close-packed $\{111\}\langle 110\rangle$ slip systems and the Peierls stress is small and weakly temperature dependent. Copper and austenitic stainless steel therefore remain ductile to cryogenic temperature and show no transition. This is the reason austenitic stainless steel is used for liquid-nitrogen and liquid-hydrogen vessels and a BCC structural steel is not.

3.4 Heat treatment: what furnace cooling, air cooling and water quenching produce

All three EN-8 specimens begin as austenite. What they become depends only on how fast they leave it:

- **Furnace cooled (annealed):** the slowest rate. Carbon has time to diffuse far, so coarse, widely spaced pearlite forms. Softest and, at room temperature, toughest.
- **Air cooled (normalised):** faster. Diffusion distance is shorter, so the pearlite is finer and the grain size smaller. Harder than annealed, by the Hall–Petch and interlamellar spacing arguments.
- **Water quenched:** fast enough to miss the nose of the transformation curve, so austenite shears to martensite — a supersaturated body-centred tetragonal solution of carbon in iron. Hardest, and brittle.

For one steel this order cannot reverse. That fact is used in Section 6.1 to assign each measured hardness to its condition.

3.5 Hardenability and the Jominy end-quench test

Hardness and hardenability are different quantities. Hardness is resistance to indentation. *Hardenability* is the depth to which a steel forms martensite under a given quench, and it is set mainly by composition: alloying elements shift the transformation curve to longer times, so martensite forms at slower cooling rates.

In the Jominy test one end of a standard bar is sprayed with water. Every distance x along the bar then experiences a different cooling rate — very fast at the quenched end, falling to roughly the rate of air cooling at the far end — so a single specimen samples the whole range. Hardness measured along a ground flat traces the structure: martensite at the quenched end, through bainite, to ferrite and pearlite far away. The resulting curve is the hardenability curve, and because the cooling rate at each distance is known and fixed by the geometry, the curve can be transferred to any other component whose cooling rate is known.

4 Procedure

1. The width and height of each Charpy specimen were measured with a vernier calliper (Table 1).
2. The specimens were soaked in the liquid nitrogen bath. Each was transferred to the machine and its temperature read with the infrared thermometer immediately before the pendulum was released, and again after fracture.
3. The pendulum was raised, the safety pin retracted, and the hammer released. The absorbed energy was read from both the electronic display and the mechanical dial. The broken halves were retained and their fracture surfaces examined and photographed (Appendix B).
4. The three EN-8 specimens were ground and three Vickers indents were made on each at 100 kgf with a 10 s dwell; both diagonals of every indent were measured.

5. The Jominy bar was austenitised, transferred to the fixture and end-quenched with the water jet until cold, then ground flat along its length to remove the decarburised layer.
6. Rockwell hardness was measured along the flat at 1.5 mm intervals to 9 mm, then at wider intervals to 50 mm. The quenched face itself was measured several times because the readings scattered.

5 Observations

Table 1: Measured Charpy specimen dimensions.

Specimen	W (mm)	H (mm)	$A_0 = (H-2)W$ (mm^2)
Cu (copper)	9.96	10.06	80.28
SS (stainless steel)	9.92	9.94	78.76
3 — air cooled	9.92	9.94	78.76
4 — water quenched	9.99	9.99	79.82
5 — furnace cooled	9.74	9.92	77.14

Table 2: Charpy impact readings. The temperature before the test is taken as the test temperature; the specimens warm during transfer from the bath.

Specimen	T_{before}	T_{after}	**Electronic** KV (J)	**Dial** KV (J)	**Used** (J)
Cu	−80 °C	−20 °C	211.72	212	212
SS	−61 °C	−21 °C	299.86[†]	107	107
3 — air cooled	−49 °C	−14 °C	4.59	3.0	3.0
4 — water quenched	−55 °C	−9 °C	4.00	2.3	2.3
5 — furnace cooled	−63 °C	0 °C	3.41	2.0	2.0

[†] *Rejected as a machine fault.* The dial reading has been taken as the reported energy throughout. Both instruments are tabulated so that the reader can see the difference, but the dial is the method the handout describes, and its quoted reading error is ±0.2 J against ±2.26 J for the electronic display. The two agree to 0.3 J on copper and to within 1.6 J on the three EN-8 specimens. For the stainless steel they do not: the display read 299.86 J against 107 J on the dial. The pendulum's capacity is 300 J, so 299.86 J is the full swing of the hammer to within 0.05 % — the display failed to register the loss rather than measuring one. A specimen that truly absorbed 299.86 J would have stopped the pendulum dead, and it did not: the stainless steel broke cleanly and the hammer carried through. 107 J is therefore used.

Table 3: Vickers hardness, 100 kgf, 10 s dwell. Three indents per specimen, two diagonals each, all in mm.

	indent 1		indent 2		indent 3		**mean** d	**HV$_{100}$**
Specimen	d_1	d_2	d_1	d_2	d_1	d_2	(mm)	
4 — water quenched	0.875	0.810	0.810	0.896	0.815	0.906	0.8520	**255.5**
3 — air cooled	1.028	1.034	1.033	1.035	1.032	1.038	1.0333	**173.7**
5 — furnace cooled	1.112	1.134	1.104	1.129	1.109	1.132	1.1200	**147.8**

Table 4: Jominy end-quench: Rockwell hardness against distance from the quenched end.

x (mm)	HRC	x (mm)	HRC	x (mm)	HRC
0 (face)	**50** (scatter 43–50)	9	24	25	17
1.5	27	10.5	22.25	30	16
3.0	26.5	12	23	35	15
4.5	26	13.5	21.5	40	14
6.0	25	15	21	45	13.5
7.5	23.75	20	19	50	13

The quenched face was measured repeatedly and the readings scattered between 43 and 50 HRC. The highest of them, **50 HRC**, is reported as the face hardness, and the rest of the spread is carried as the error bar in Figure 3. The highest reading was taken rather than the mean, for the following reason. The quenched face is the one place on the bar where the cooling rate is certainly fast enough for a fully martensitic structure, and the hardness of fully martensitic plain carbon steel is fixed almost entirely by its carbon content: for EN-8, at about 0.4 wt% C, that hardness is of the order of 55–57 HRC. Every mechanism that could disturb a reading on this particular surface lowers it, and none raises it:

- the end face is the surface the water jet struck directly, so it carries scale and a decarburised skin that the ground flat does not;
- it is not flat or polished to the standard the Rockwell test assumes, and a face that is not square to the indenter always reads *low*, never high;
- it is the face the bar stood on in the fixture.

ASTM A255 accordingly begins its traverse at $1/16''$ (1.6 mm) rather than at zero. The 0 mm point is reported here for completeness, with its full scatter shown, and is excluded from any fit.

6 Results and Analysis — Individual Group Activity

6.1 Assigning each hardness to its heat treatment

The three EN-8 specimens were indented in no fixed order. The assignment nevertheless follows without ambiguity from two independent records. First, the heat-treatment key written in the notebook (Appendix B): specimen 3 air cooled, specimen 4 water quenched, specimen 5 furnace cooled. Second, for a single steel the order of hardness cannot reverse: furnace cooled is always softest, air cooled intermediate, water quenched hardest. Sorting the three measured hardnesses therefore assigns each one uniquely, and the two records agree:

Specimen	HV_{100}	Condition	Expected structure
4	255.5	water quenched	martensite + bainite
3	173.7	air cooled	fine pearlite + ferrite
5	147.8	furnace cooled	coarse pearlite + ferrite

A third, independent check is available in this experiment. The water-quenched specimen reads 255.5 *HV*, which converts to approximately 23 HRC. The Jominy bar of the *same* steel reads 23 HRC at about 12 mm from the quenched end — a cooling rate comparable to that experienced by the interior of a 10 mm square bar quenched in water. The two experiments agree.

6.2 Specific impact energy

Using Equation 2 with the dial energies and the measured areas of Table 1:

Table 5: Specific impact energy at the measured test temperatures. This is the table requested in the handout.

Material	Temperature	KV (J)	a_k (J/cm^2)	a_k (J m^{-2})	HV_{100}
Cu	−80 °C	212	264.1	2.64×10^6	—
SS (stainless steel)	−61 °C	107	135.9	1.36×10^6	—
EN-8 air cooled	−49 °C	3.0	3.81	3.81×10^4	173.7
EN-8 water quenched	−55 °C	2.3	2.88	2.88×10^4	255.5
EN-8 furnace cooled	−63 °C	2.0	2.59	2.59×10^4	147.8

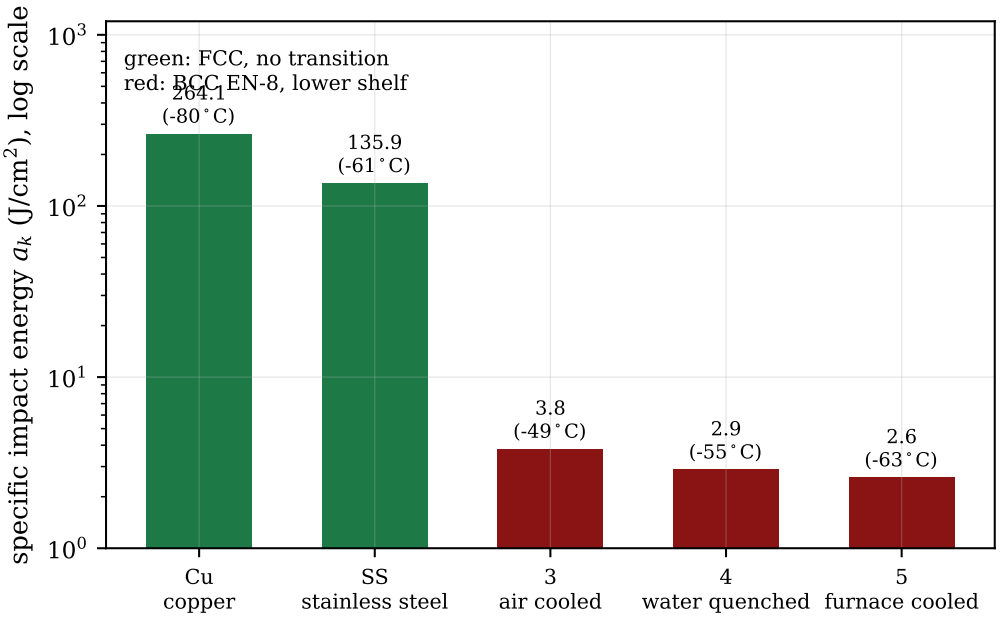


Figure 1: Specific impact energy of all five specimens on a logarithmic scale. The two face-centred cubic materials absorb between forty and a hundred times more energy than any of the three body-centred cubic EN-8 conditions at the same kind of temperature.

The size of this separation is the main result. Copper at −80 °C absorbed 212 J and the stainless steel at −61 °C absorbed 107 J, while all three EN-8 specimens absorbed between 2.0 and 3.0 J. The two FCC materials are on their upper shelf — they have no transition to be on the far side of. The three BCC specimens are all on their lower shelf. Heat treatment changes the hardness of EN-8 by a factor of 1.7, but at these temperatures it does not rescue the toughness of any of the three: every one of them is fully brittle.

6.3 Impact toughness against Vickers hardness

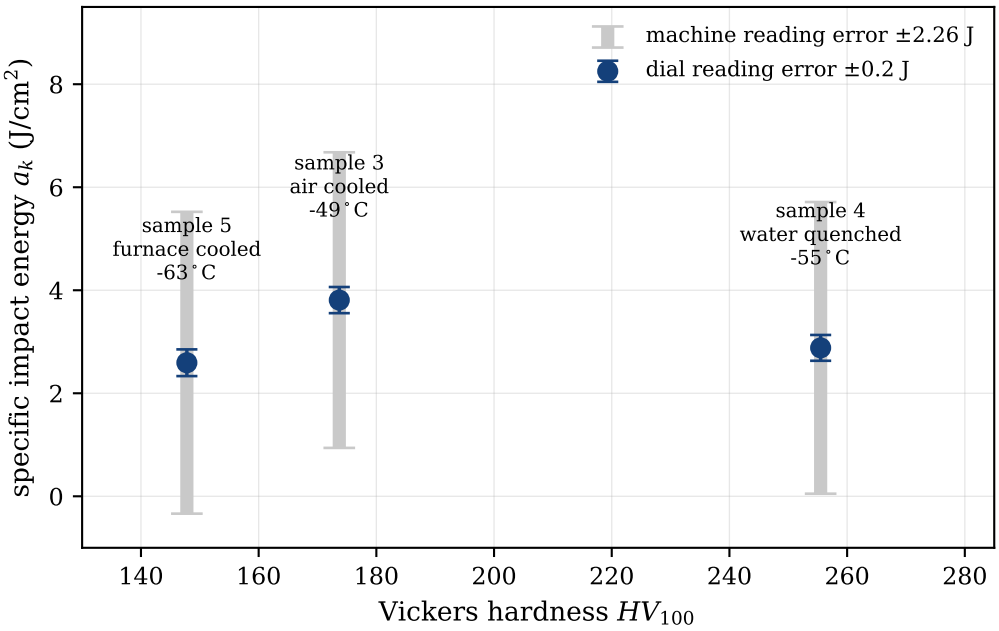


Figure 2: Specific impact energy against Vickers hardness for the three EN-8 conditions. The narrow bars are the dial error, ±0.2 J; the wide grey bars are the electronic display error, ±2.26 J. Each specimen was tested at a different temperature, given beside its point.

The expected result is a falling line: the harder the steel, the less energy it absorbs. The data is weaker than that, and the limits of what it can support are set out below.

- On the dial error alone, the air-cooled specimen (3.81 J/cm^2) is separated from the water-quenched (2.88) and the furnace-cooled (2.59) by about four standard errors. Those separations are real.
- The water-quenched and furnace-cooled specimens differ by 0.29 J/cm^2, about one standard error. They cannot be told apart.
- If the electronic reading error of ±2.26 J is the honest figure for the measurement, *none* of the three can be separated.
- The three were tested at −49, −55 and −63 °C. On the lower shelf the energy varies only slowly with temperature, but the specimen with the highest energy is also the one tested warmest, so treatment and temperature are confounded in these three points.

The physically expected ordering — furnace cooled toughest, water quenched most brittle — is *not* reproduced: the furnace-cooled specimen gave the lowest energy of the three. This does not contradict the theory. Three single measurements of 2–3 J, taken at three different temperatures on a machine whose own quoted error is the same size as the measurements themselves, cannot resolve an ordering. Separating the three conditions would require the test to be repeated at several temperatures on each condition.

6.4 Fracture surfaces

The broken specimens were photographed together after the test (Appendix B, panels **e** and **f**). The five fall into three visibly distinct classes, and the classes follow the energies exactly.

Copper — 212 J, fully ductile, did not separate. The copper specimen *never broke into two pieces*. The pendulum drove it through the anvil gap and folded it into a V, leaving the two arms still joined by a thin ligament at the notch. The notch root is drawn down into a deep, narrow waist: the sides have pulled visibly inward, which is lateral contraction, and the matching back face has bulged out. What can be seen of the torn material at the root is dull, rough and fibrous, with a torn-metal lustre rather than any facets, and the edges carry heavy burred shear lips where the material was sheared over the anvil shoulders. No crystalline area is visible anywhere. Since the specimen never separated, no shear-fracture percentage can properly be quoted for it, and the 212 J must be read with care: it is the work of bending and tearing a specimen the hammer could not break, so it is a lower bound on the copper's toughness rather than a Charpy fracture energy. Copper is face-centred cubic and has no transition; at −80 °C it is still on its upper shelf.

Stainless steel — 107 J, ductile, separated. The stainless steel did break into two halves. Its fracture faces are dull and matte, with a fine even texture under a slightly bluish cast from the lighting — the appearance of microvoid coalescence, not of cleavage. Curved ridges sweep across each face in a chevron pattern that points back towards the notch, marking the direction in which the crack ran, and the edges carry clear shear lips. There is visible lateral contraction at the notch. Austenitic stainless steel is also face-centred cubic and also has no transition; it absorbed half the copper's energy simply because it is a stronger material that tears at a higher load over a shorter distance.

The three EN-8 specimens — 2–3 J, fully brittle, all six halves alike. All three EN-8 specimens snapped cleanly into two halves. Every one of the six fracture faces is *flat and shiny*: square to the bar axis, with no lip, no curl and no measurable lateral expansion at the sides, and with the bright, sparkling, granular look of cleavage facets catching the light. This is 100 % cleavage and 0 % shear. The three cannot be told apart by eye: furnace cooled, air cooled and water quenched all produced the same flat crystalline face. Some of the faces carry brown oxide staining and dull patches, which is condensation and rust picked up after the specimens came out of the liquid nitrogen bath; the underlying texture beneath the staining is faceted in every case.

What the surfaces add to the energies. The surfaces agree with the energies, and they were obtained independently of them. The two face-centred cubic materials gave 107 and 212 J and show fibrous surfaces with shear lips, or no separation at all; the three body-centred cubic specimens gave 2.0 to 3.0 J and show flat cleavage.

The surfaces also settle a question the energies could not. Section 6.3 showed that the three EN-8 conditions cannot be separated by their impact energies within the machine error, and left open whether that was a limitation of the measurement or a real physical result. The fracture surfaces answer it: all six halves are fully cleaved, so at liquid-nitrogen soaking temperature all three conditions are on the lower shelf, where the fracture mode is cleavage whether the structure is coarse pearlite, fine pearlite or martensite. There is nothing for the heat treatment to change. The shear-fracture percentage, the standard index of position on the transition curve, is 0 % for all three.

7 Results and Analysis — Jominy End-Quench

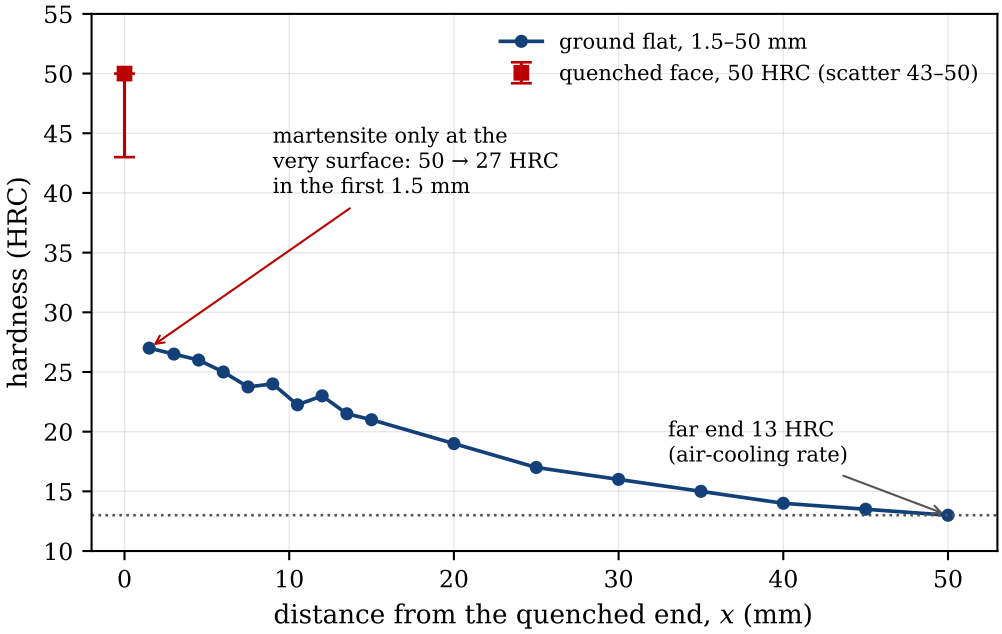


Figure 3: Hardenability curve for the EN-8 bar. The square point at $x = 0$ is the quenched face: the highest of the repeat readings, with the full 43–50 HRC scatter as its error bar. That face lies off the ground flat and is excluded from the trend.

7.1 How the hardness varies with distance

The curve has three parts. Over the first 1.5 mm the hardness falls steeply, from a face value of 50 HRC to 27 HRC. From 1.5 to about 15 mm it declines gently and almost linearly, 27 → 21 HRC, with scatter of about ±1 HRC between neighbouring points — the 7.5 and 9 mm readings (23.75 and 24) and the 10.5 and 12 mm readings (22.25 and 23) each invert the trend by less than the reproducibility of the instrument. Beyond 20 mm the curve flattens towards a plateau, reaching 13 HRC at 50 mm and changing by only 2 HRC over the last 15 mm.

This is the shape expected of a steel of low hardenability. The martensite is confined to a thin surface layer; by a few millimetres in, the cooling rate has already fallen below what is needed to miss the transformation nose, and the structure is bainitic and then pearlitic.

One discrepancy, and what it means. The step between the face and the first point on the ground flat is very large: 50 HRC at 0 mm against 27 HRC at 1.5 mm, a fall of 23 HRC in a millimetre and a half. The handout's own micrographs, taken previously by the teaching assistants on EN-8, show a fully martensitic structure 1 mm from the quenched end, and a fully martensitic 0.4 % carbon steel should read in the mid-fifties on the C scale — not 27. Three explanations are possible, and they are not exclusive:

1. **The flat was ground too deep.** The specification is 0.38 mm, enough to take off scale and decarburised metal. Grind further and the hardened skin goes with it, and the first reading on the flat is taken on material that cooled more slowly than the surface did. The abruptness of the step points to this as the main cause.
2. **Slow transfer from the furnace.** Any delay between the furnace and the water jet lets the surface begin to transform before the quench starts.

3. **EN-8 really has very little hardenability.** The handout's micrographs were taken at 850 °C on a different bar; a plain carbon steel with no deliberate alloying has an extremely steep hardenability curve, and a fall of this kind over the first few millimetres is the expected shape, if not the expected magnitude.

The Charpy half of this experiment supports the third. The water-quenched 10 mm Charpy bar — the same steel, given the fastest quench available — reached only 255.5 HV ($\approx$ 23 HRC), nowhere near a martensitic hardness either. Two independent measurements therefore agree that water-quenched EN-8 reaches only 23–27 HRC a millimetre or two below the surface. That is the main quantitative result about this steel: it does not through-harden.

7.2 Where the cooling rate equals air cooling

The far end of a Jominy bar cools essentially by conduction into its own length and by convection to still air, and is conventionally taken as the air-cooled condition. The data supports this directly, because the air-cooled Charpy specimen of the same steel was measured independently: 173.7 HV_{100}. Converting the far-end Jominy readings, 13–15 HRC corresponds to roughly 190–210 HV, within about 15 % of the air-cooled specimen and above it.

On this basis the cooling rate becomes equivalent to air cooling from about **35–40** mm onward, where the curve has flattened to within 2 HRC of its end value. **This comparison carries a caveat:** hardness below 20 HRC is outside the useful range of the Rockwell C scale, which is why ASTM E18 directs such material to the B scale, so both the readings and their conversion to HV are approximate at the far end of this bar. The traverse was taken on the C scale throughout, as the handout directs.

7.3 The Jominy distance for 50 % martensite

Hardenability is defined as the distance at which the structure is 50 % martensite, and fixing it properly requires metallography, which was not performed in this lab. The handout supplies micrographs taken previously by the teaching assistants for EN-8 austenitised at 850 °C for 30 minutes: fully martensitic at 1 mm, and fully ferritic and pearlitic by 50 mm, with the change occurring over the first few millimetres. The 50 % point therefore lies between 2.5 and 5 mm on those micrographs.

The hardness data measured here places it earlier. The standard correlation between carbon content and the hardness of a half-martensitic structure — the curve reproduced in every textbook treatment of hardenability, for example Krauss, *Steels: Processing, Structure and Performance* — puts the 50 % martensite hardness of a 0.4 wt% carbon steel at roughly 45 HRC, against about 55–57 HRC for the same steel fully martensitic. Only the quenched face of this bar, at 50 HRC, lies above the 45 HRC line; the first point on the ground flat is already at 27 HRC. On the hardness criterion alone, therefore, the 50 % martensite point in this bar lies inside the first millimetre:

$$\boxed{J_{50\%} \lesssim 1 \text{ mm}}$$

This is shallower than the handout's micrographs suggest, for the reasons set out above, and it should be read as an upper bound on the hardenability rather than a precise figure — a proper determination needs the metallography that was not performed here. Either way the conclusion is the same and is not sensitive to the exact number: EN-8 forms half martensite only within about a millimetre of a water-quenched surface.

8 Note on the across-group activity

This report covers the *individual group activity* set out in the handout, in full. The *activity across different groups* — the impact energies of all eight materials compared at the three test

temperatures, and the hardenability curves of all six batches on one plot with the effect of composition and austenitising temperature — will be submitted as a separate report for this experiment, once the class data has been collected.

The split is unavoidable. Group 1 Friday tested at liquid nitrogen temperature only, so no transition temperature, no 27 J index and no 50 % FATT can be obtained from this group's data alone; a transition curve needs the three test temperatures that the handout distributes across the batches. Nothing in the present report depends on that data, and nothing in it will change when the comparison is made.

9 Discussion

The experiment separates two ideas that are easily confused: how hard a steel is, and how deeply it can be hardened.

The Charpy results show that hardness and toughness are not two names for the same thing, and also that the relation between them is not simple. Across the three EN-8 conditions the hardness spans 147.8 to 255.5 HV, a factor of 1.7, while the impact energy spans 2.0 to 3.0 J — a difference that the machine can barely resolve. At liquid-nitrogen soaking temperatures the steel is on its lower shelf whatever its heat treatment, and on the lower shelf the energy is governed by the cleavage stress, which is not much altered by the difference between coarse pearlite, fine pearlite and martensite. The heat treatment would separate these three specimens clearly at room temperature; at $-50\,^{\circ}\mathrm{C}$ it does not.

The contrast with copper and the stainless steel is the opposite case, and it is large enough that no error analysis is needed to see it: a factor of between forty and a hundred in specific energy. That gap comes from crystal structure, not from strength. Twelve close-packed slip systems with a low Peierls barrier remain available at any temperature reached here, so plastic work continues to absorb energy; in the body-centred cubic lattice it does not.

The Jominy curve adds the third idea. The same water quench that produced 255.5 HV in a 10 mm Charpy bar produced 50 HRC at the face of the Jominy bar and only 27 HRC 1.5 mm in. Hardenability is a property of composition, and EN-8, with no deliberate alloying, has little of it. An alloy steel such as EN-24, carrying nickel, chromium and molybdenum, holds its hardness far along the bar because those elements retard the diffusive transformation and let martensite form at much slower cooling rates. That comparison belongs to the across-group activity, submitted separately, and it is why the two steels were distributed between the Monday and Friday batches.

There is a practical consequence. Choosing EN-8 for a component thicker than a few millimetres and expecting a through-hardened part would fail: the surface would be hard and the core soft. The Jominy curve is the quantitative statement of that limit.

10 Conclusions

1. At liquid-nitrogen soaking temperature all three EN-8 conditions lie on the lower shelf, absorbing 2.0–3.0 J (2.59–3.81 J/cm^2), while copper absorbed 212 J at $-80\,^{\circ}\mathrm{C}$ and the stainless steel 107 J at $-61\,^{\circ}\mathrm{C}$. The face-centred cubic materials show no ductile-to-brittle transition; the body-centred cubic steel does.
2. Vickers hardness at 100 kgf gave 255.5 HV water quenched, 173.7 HV air cooled and 147.8 HV furnace cooled, in the order required by the three cooling rates, and confirming the heat-treatment key recorded in the laboratory.
3. Impact energy did not fall monotonically with hardness. Within the dial error the air cooled specimen is separated from the other two, but the water-quenched and furnace-cooled specimens are not separable, and within the electronic reading error none of the

three are. Three single measurements at three different temperatures on the lower shelf are not enough to establish the trend.

4. The Jominy bar fell from 50 HRC at the quenched face to 27 HRC at 1.5 mm and 13 HRC at 50 mm, the signature of a plain carbon steel of low hardenability. The cooling rate becomes equivalent to air cooling beyond about 35–40 mm.
5. The fracture surfaces confirm the energies independently. All six EN-8 halves broke flat and shiny — 100 % cleavage, 0 % shear, no lateral expansion and no shear lips — with the three heat treatments indistinguishable by eye. The stainless steel broke with a dull fibrous face, chevron markings and shear lips. The copper did not break at all: it folded into a V with a deep waist at the notch, so its 212 J is a lower bound on its toughness.
6. The water-quenched Charpy specimen (255.5 $HV \approx 23$ HRC) matches the Jominy bar at about 12 mm, an independent and self-consistent link between the two halves of this experiment. Both say the same thing: EN-8 does not through-harden, and a millimetre or two below a water-quenched surface it is already soft.

A Sample calculations

Vickers hardness, water-quenched specimen. The mean of the six diagonals is

$$d = \tfrac{1}{6}(0.875 + 0.810 + 0.810 + 0.896 + 0.815 + 0.906) = 0.8520 \text{ mm},$$

and with $F = 100$ kgf,

$$HV = \frac{1.8544\,F}{d^2} = \frac{1.8544 \times 100}{0.8520^2} = \frac{185.44}{0.72590} = 255.5.$$

Effective area, water-quenched specimen.

$$A_0 = (H - 2)\,W = (9.99 - 2) \times 9.99 = 7.99 \times 9.99 = 79.82 \text{ mm}^2 = 0.7982 \text{ cm}^2.$$

Specific impact energy, water-quenched specimen.

$$a_k = \frac{KV}{A_0} = \frac{2.3 \text{ J}}{0.7982 \text{ cm}^2} = 2.88 \text{ J/cm}^2 = 2.88 \times 10^4 \text{ J m}^{-2}.$$

Propagated error on a_k. With the dial error ± 0.2 J, $\delta a_k = 0.2/0.7982 = 0.25$ J/cm^2; with the display error ± 2.26 J, $\delta a_k = 2.26/0.7982 = 2.83$ J/cm^2. The second is as large as the measurements themselves, which is the basis of the caution in Section 6.3.

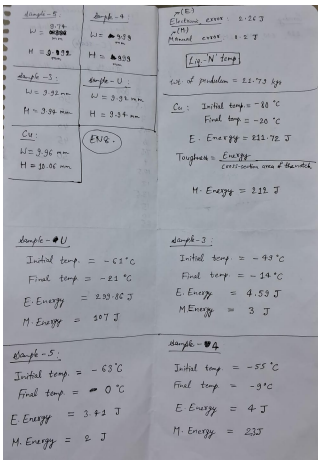
Hardness at the quenched face: why 50 HRC and not the mean. The repeat readings scattered over 43–50 HRC, far wider than the ± 1 HRC reproducibility seen along the ground flat. Averaging is correct only when the error is random about a true value, and here it is not: scale, a decarburised skin and a face that is neither ground nor square to the indenter all drive a reading *downwards*, and nothing drives it up. The distribution is one-sided, so the estimator of the underlying hardness is the maximum:

$$H_{\text{face}} = \max\{43,\ \ldots,\ 50\} = 50 \text{ HRC}, \qquad \text{scatter } 43\text{–}50 \text{ HRC (reported in full)}.$$

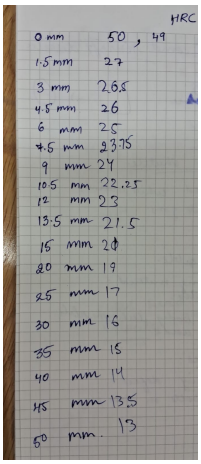
The expected hardness of fully martensitic EN-8 (55–57 HRC at 0.4 wt% C) agrees. Nothing downstream turns on the choice: ASTM A255 does not measure at $x = 0$ at all, so the point is excluded from every fit.

B The original record: notebook and specimens

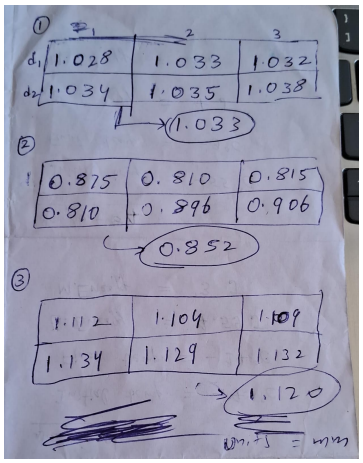
The primary record: notebook pages written in the laboratory on 30 October 2026, and the broken specimens photographed on 13 November 2026. Every number read off them is in Section 5.



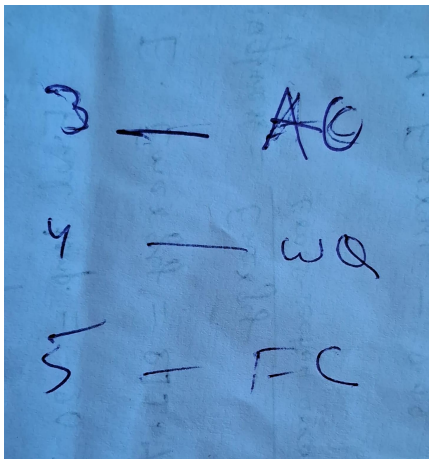
(a) Charpy: specimen dimensions, machine errors, test temperatures, energies.



(b) Jominy: Rockwell C against distance from the quenched end.



(c) Vickers: three indents per specimen, two diagonals each, in mm.



(d) The heat-treatment key: 3 air cooled, 4 water quenched, 5 furnace cooled.



(e) All five after test. Left: copper, bent into a V, still in one piece. Centre: six EN-8 halves, flat and faceted. Bottom: stainless steel, dull and fibrous.



(f) Closer. The granular EN-8 faces are cleavage; the copper has torn, not cleaved; the stainless half at right shows chevrons and a shear lip.